



Other Types of Respiratory Failure

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Respiratory failure can occur perioperatively or result from hypoperfusion.

Perioperative respiratory failure is usually caused by [atelectasis](#) (collapse of lung tissue with loss of volume), and increases the risk of pneumonia ([1](#)). (See also [Overview of Mechanical Ventilation](#).)

Atelectasis caused by abdominal distention should be alleviated by treating the cause (eg, nasogastric suction for excessive intraluminal air, paracentesis to evacuate tense ascites).

General strategies for preventing or treating atelectasis include ([2](#)):

- Incentive spirometry
- Ensuring adequate analgesia for chest and abdominal incisions
- Upright positioning
- Early mobilization

Incentive spirometry is widely used but ineffective alone ([3](#), [4](#), [5](#)).

Hypoperfusion, regardless of cause, may result in respiratory failure through inadequate delivery of oxygen to respiratory muscles coupled with excess respiratory muscle load (eg, acidosis, [sepsis](#)). Mechanical ventilation can divert blood flow from overworked respiratory muscles to critical organs such as the brain, kidneys, and gut.

References

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